ELECTRIC VEHICLE POPULATION DATA ANALYSIS USING PYTHON

Project Documentation Report

Dataset	Data.gov — Electric Vehicle Population Data
Location	Washington State, USA
Domain	Electric Vehicles / Transportation
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab
Analysis Type	Statistical Analysis & Data Visualization
Repository	*https://github.com/AtchayaChandran/Electric-Vehicle-Population-Data-Analysis*

TABLE OF CONTENTS

Project Objective

The main objective of this project is to analyze the **Electric Vehicle Population Data** using Python and identify meaningful patterns and insights related to electric vehicle adoption.

Objectives

- To analyze the distribution of different **Electric Vehicle Types**.
- To identify the **top EV manufacturers**.
- To study the distribution of EVs across different **Model Years**.
- To analyze the **Electric Range** of vehicles.
- To identify the **counties with the highest EV population**.
- To analyze **CAFV Eligibility**.
- To compare relationships between **EV Type, Electric Range, Model Year, Manufacturer, and County** using statistical analysis and visualizations.

Dataset Information

Source	Data.gov — Electric Vehicle Population Data
Records / Columns	289,564 rows × 16 columns
Location	Washington State, USA
Domain	Electric Vehicles / Transportation
Data Type	Electric Vehicle registration data
Key Variables	EV Type, Make, Model, Model Year, Electric Range, County, City, CAFV Eligibility

Key Colums & Initial Data Inspection

Key Columns	**EV Type, Make, Model, Model Year, Electric Range, County, City, CAFV Eligibility**
Initial Data Inspection	**Checked the first and last records, dataset shape, column names, data types, statistical summary, unique values, duplicate records, and missing values.**

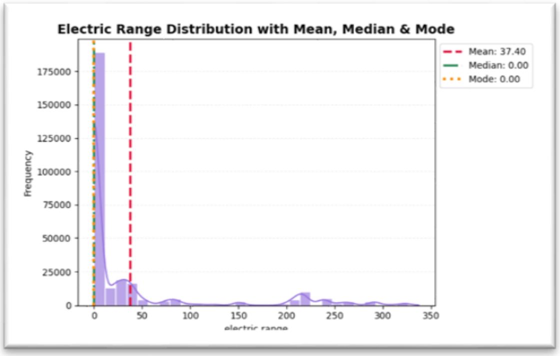
Data Cleaning & Feature Engineering

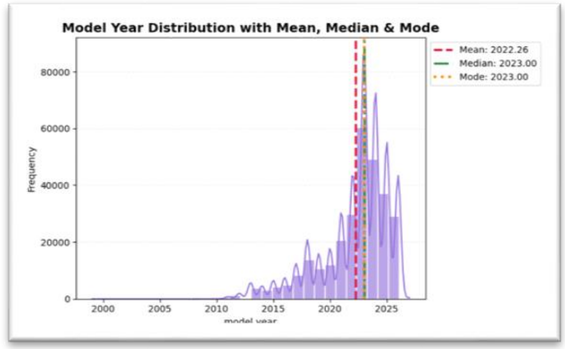
Data Cleaning	Checked and handled missing values & duplicate records
Data Type Conversion	Converted columns into appropriate data types for analysis.
Column Cleaning	Removed unnecessary columns and renamed columns where required.
Feature Engineering	Created useful derived features for further analysis.

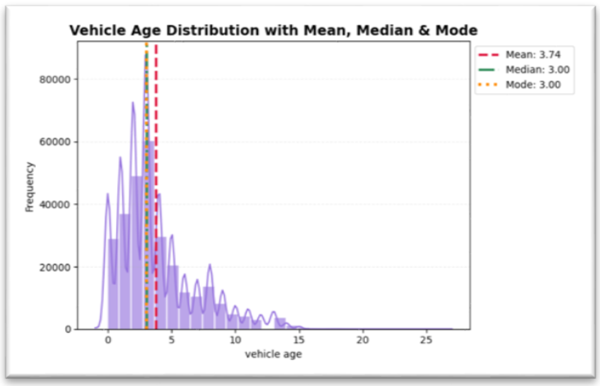
4. Statistical Analysis

4.1 Measures of Central Tendency (Mean, Median, Mode)

- Electric Range Distribution
- Model Year Distribution
- Vehicle Age Distribution



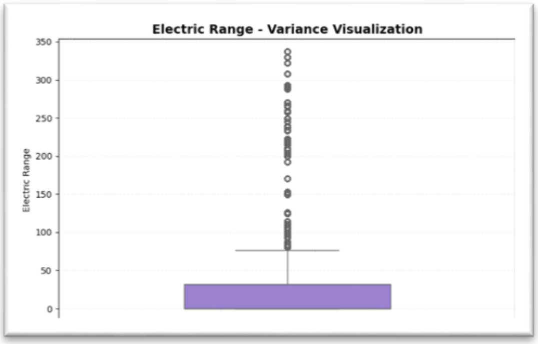


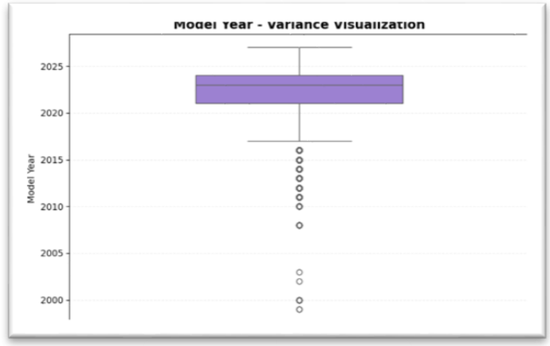


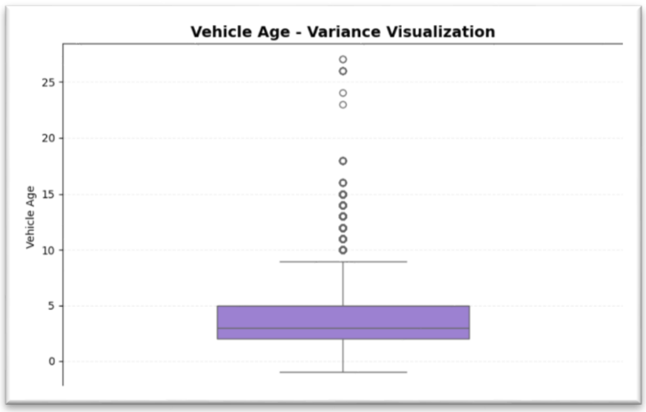
Interpretation: The central tendency analysis shows that the dataset is mainly concentrated around **recent model years and relatively newer vehicles**. Electric Range has a **mean of 37.40 miles**, while its median and mode are **0 miles**, indicating that many vehicles have zero or very low electric range. The Model Year has a mean of **2022.26** and a median and mode of **2023**, while Vehicle Age has a mean of **3.74 years** and a median and mode of **3 years**. Overall, the results indicate that the dataset contains predominantly **recent and relatively new electric vehicles**, with Electric Range showing a more uneven distribution.

4.2 Measures of Dispersion (Variance)

- Electric Range – Variance Visualization
- Model Year – Variance Visualization
- Vehicle Age – Variance Visualization



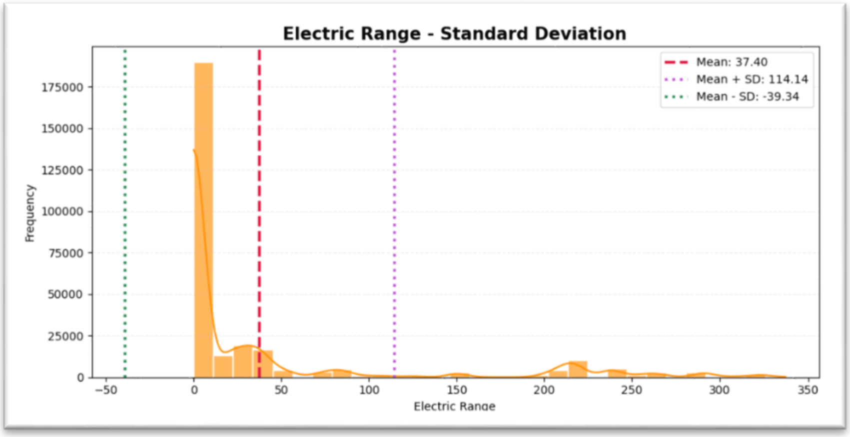


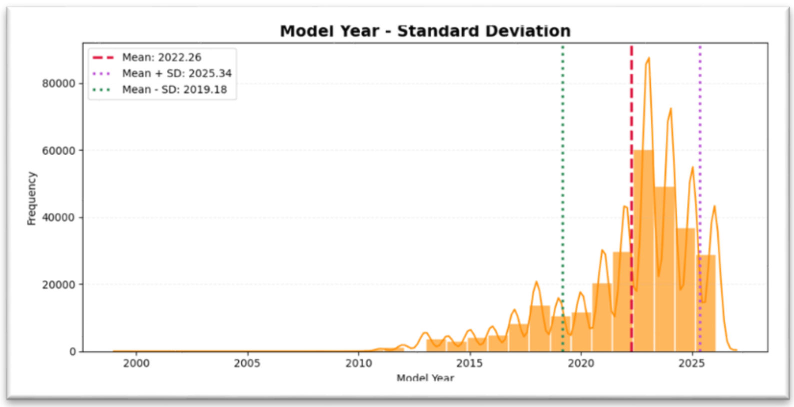


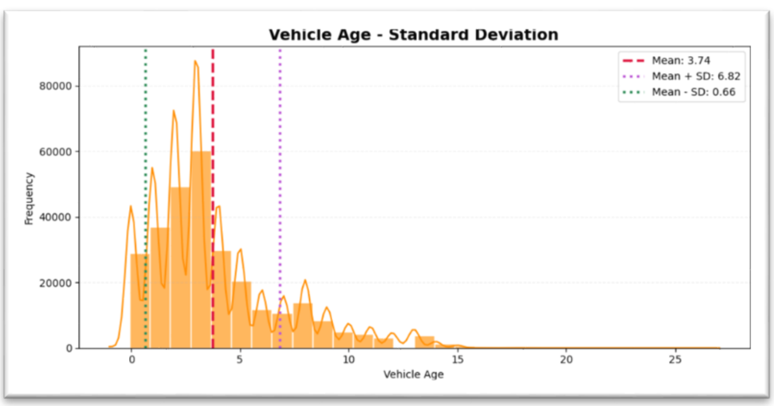
Interpretation: The variance analysis shows that **Electric Range has the highest variation**, with a variance of **5889.14**, indicating considerable differences in electric driving range among the vehicles. Model Year has a much lower variance of **9.49**, showing that most vehicles are concentrated around recent model years. Overall, Electric Range shows much greater variability than Model Year and Vehicle Age.

Standard Deviation

- Electric Range – Standard Deviation
- Model Year – Standard Deviation
- Vehicle Age – Standard Deviation



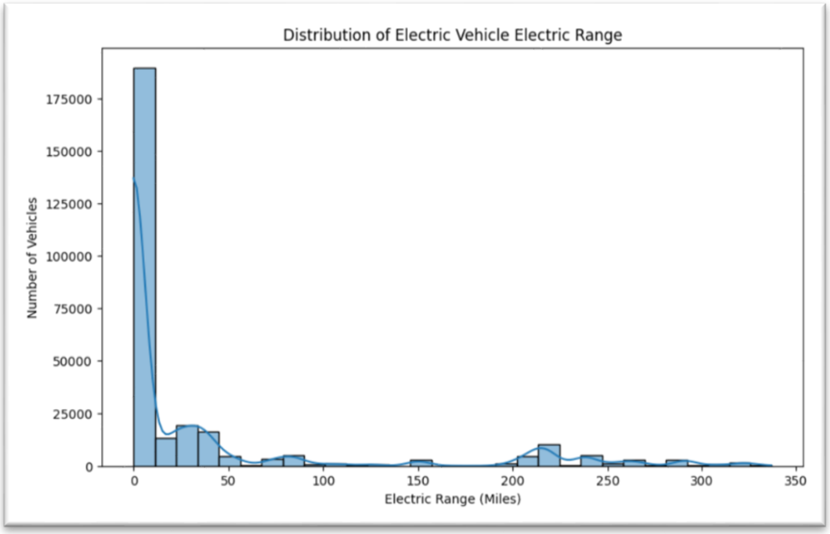




Interpretation: The standard deviation analysis shows that **Electric Range has the greatest variability**, with a standard deviation of **76.74 miles**, compared with its mean of **37.40 miles**. Model Year and Vehicle Age both have a standard deviation of **3.08 years**, showing that these values are more concentrated around their respective means.

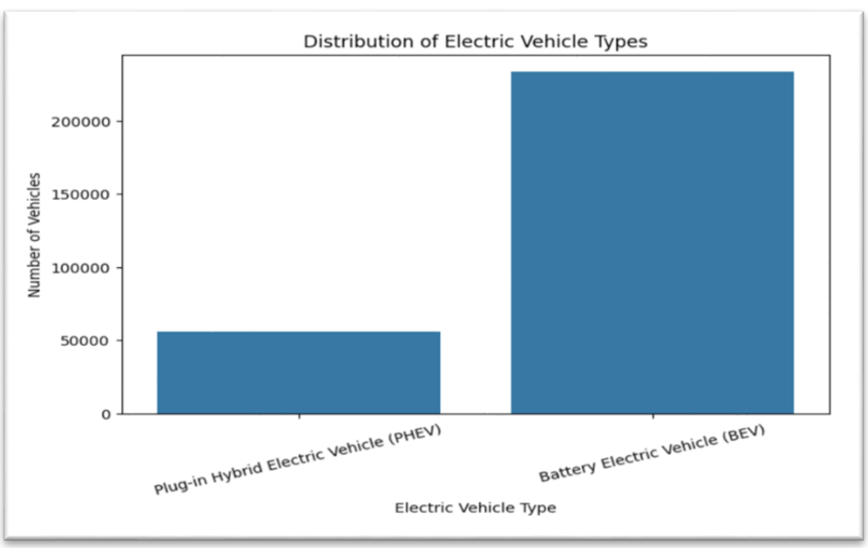
5. Univariate Analysis

1. Distribution of Electric Vehicle Range



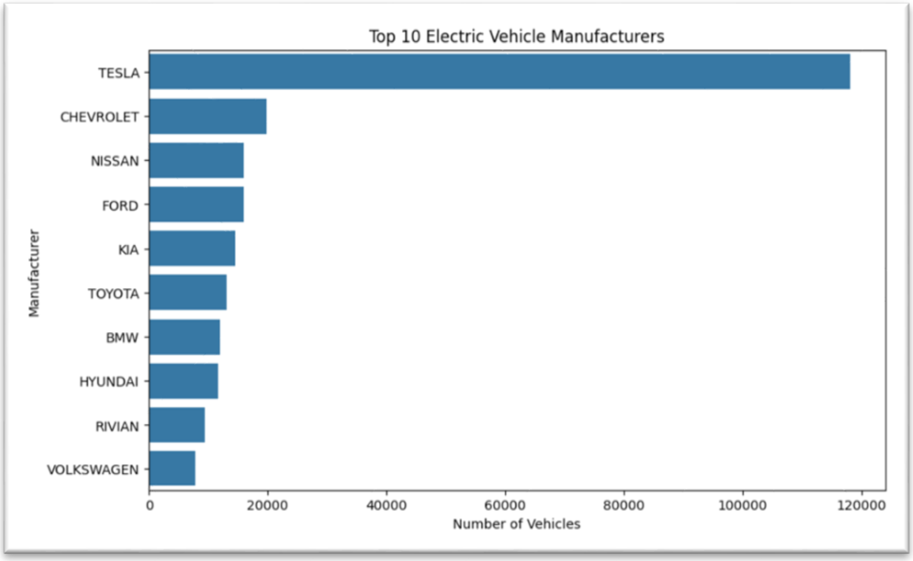
Electric Range values are highly concentrated around 0 miles, with a smaller number of vehicles having higher electric ranges, indicating an uneven distribution of electric driving range.

2. Distribution of Electric Vehicle Types



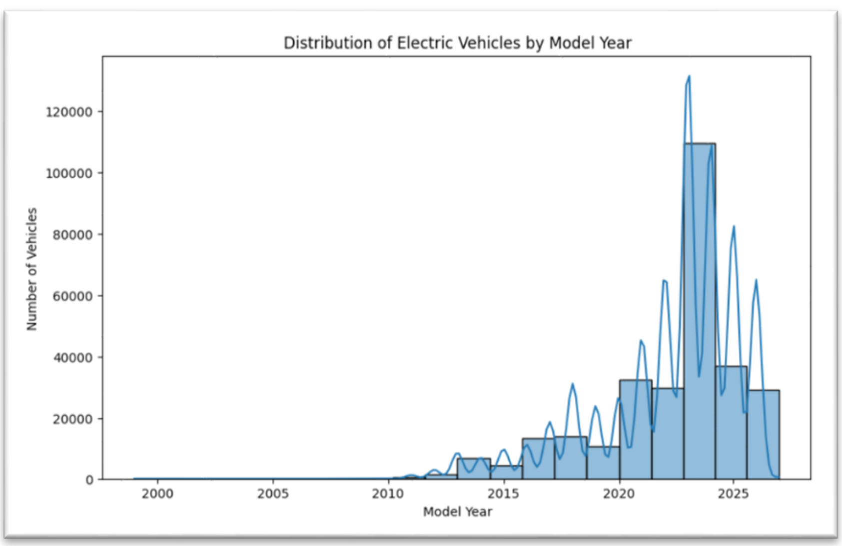
Battery Electric Vehicles (BEVs) have a much higher representation than Plug-in Hybrid Electric Vehicles (PHEVs), showing that BEVs form the majority of vehicles in the dataset.

3. Top 10 Electric Vehicle Manufacturers



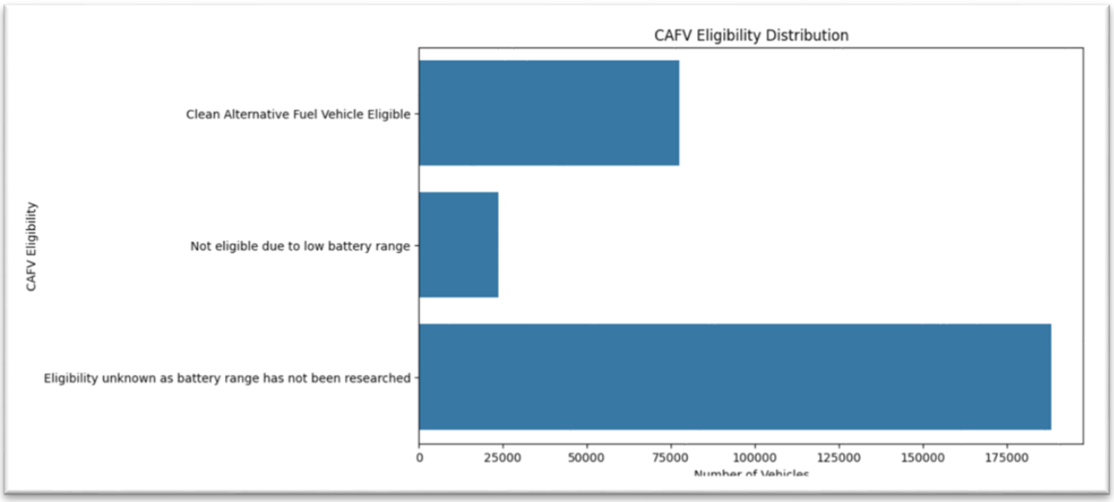
Tesla has the highest representation among the manufacturers, followed by Chevrolet, Nissan, Ford, and Kia, showing that EV registrations are concentrated among a few major manufacturers.

4. Model Year Distribution



Recent model years have stronger representation in the dataset, with newer electric vehicles making up a significant portion of the registered vehicles.

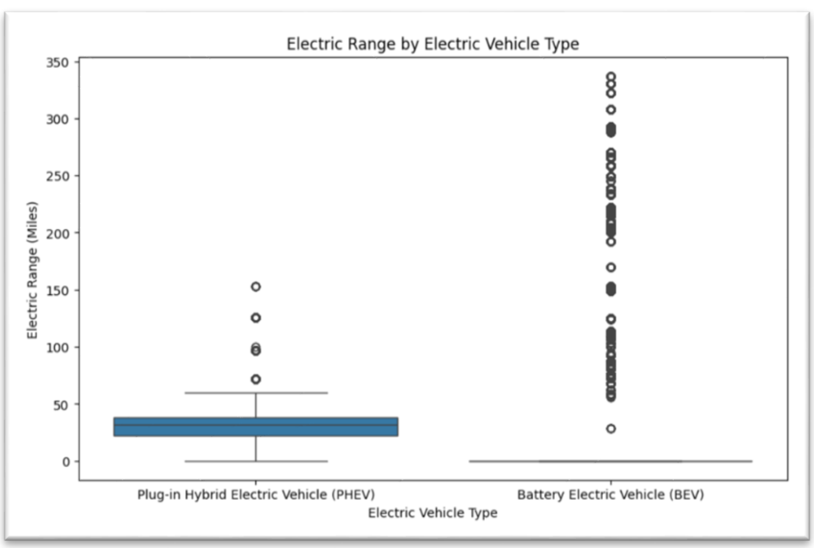
5. CAFV Eligibility Distribution



The vehicles are distributed across different CAFV eligibility categories, with a large proportion of records having an unknown eligibility status, while 77,580 vehicles are classified as CAFV eligible.

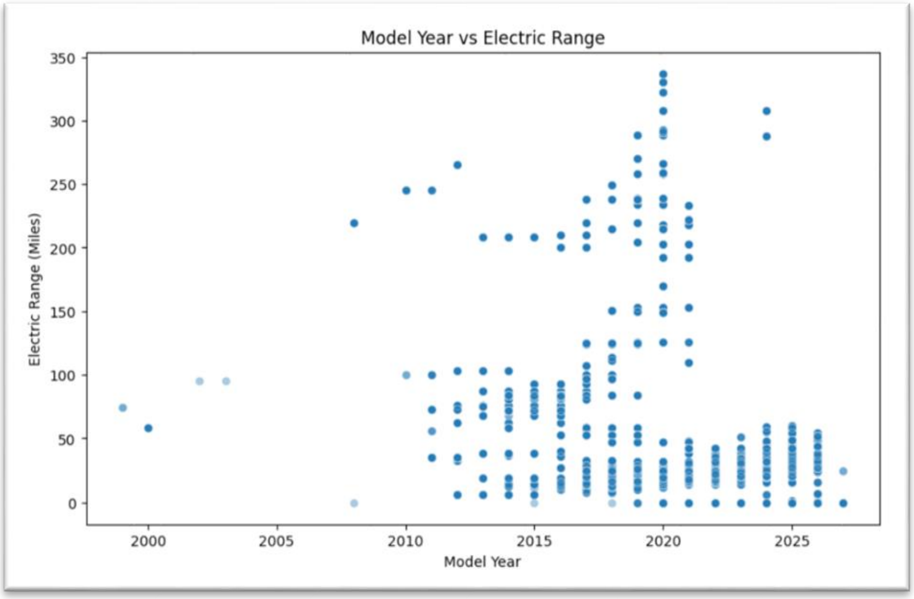
5.2 Bivariate Analysis

1. EV Type vs Electric Range



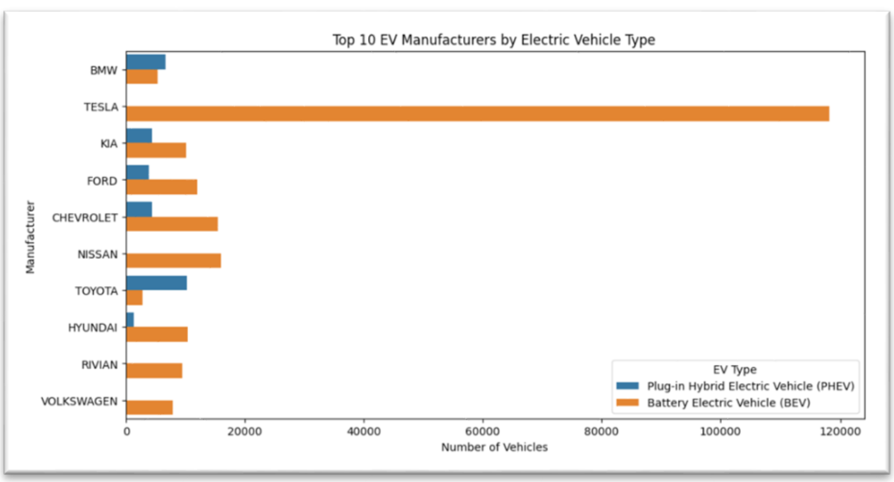
BEVs and PHEVs show different Electric Range characteristics. BEVs generally have a wider range distribution, while PHEVs have a comparatively smaller electric-only range, indicating that EV type is related to Electric Range.

2. Model Year vs Electric Range



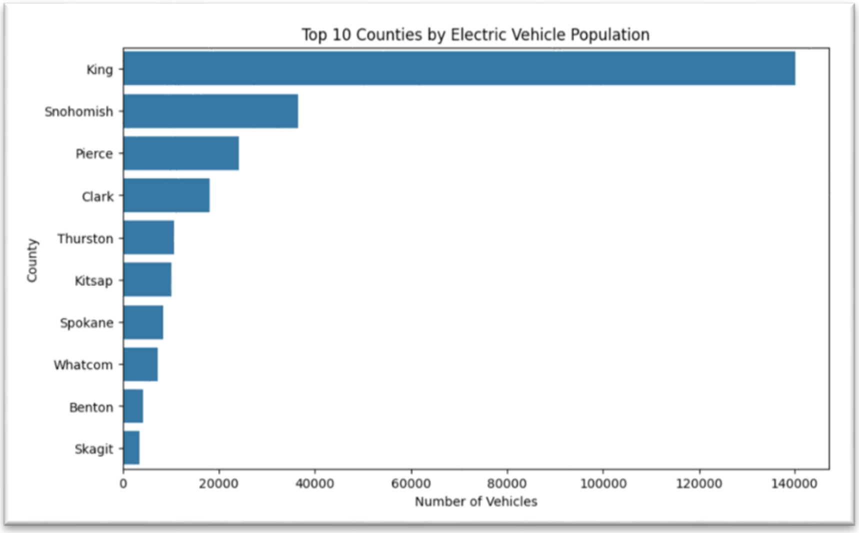
Electric Range varies considerably across different model years. More recent model years show a stronger presence of higher Electric Range values compared with many older model years, suggesting improvements in electric vehicle range over time.

3. Next Visual — EV Manufacturer vs Electric Range



The visualization compares Electric Range across different EV manufacturers. The variation in range across manufacturers indicates differences in vehicle technology and model capabilities.

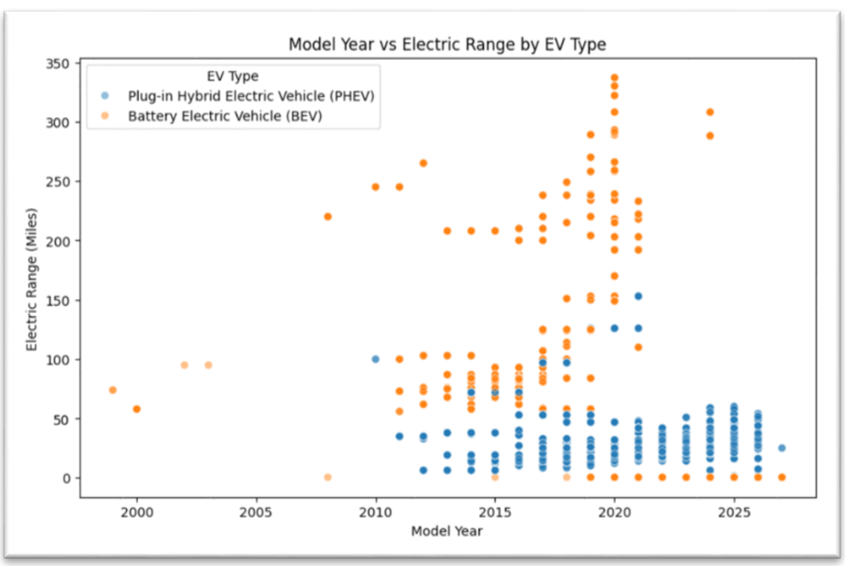
4. Top 10 Counties by Electric Vehicle Population



EV adoption is geographically concentrated rather than evenly distributed across counties. King County has the highest EV population, followed by Snohomish and Pierce counties.

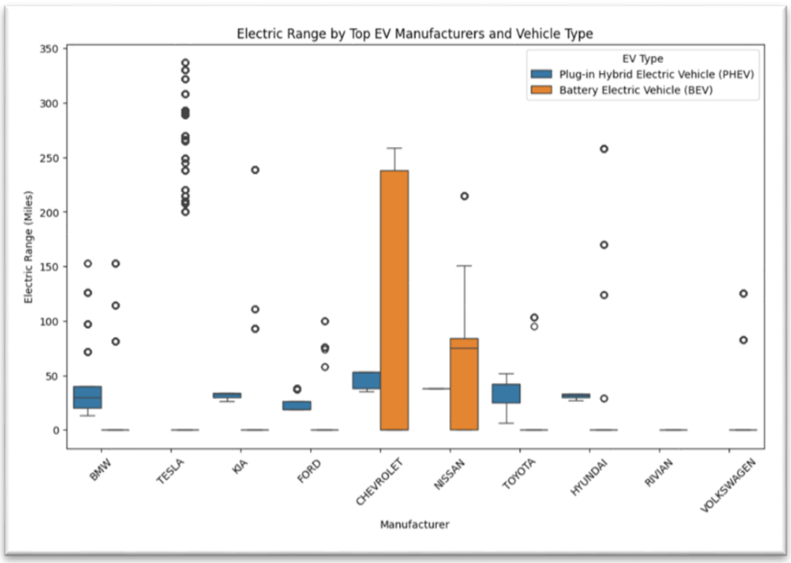
5.3 Multivariate Analysis

1. Model Year vs Electric Range by EV Type



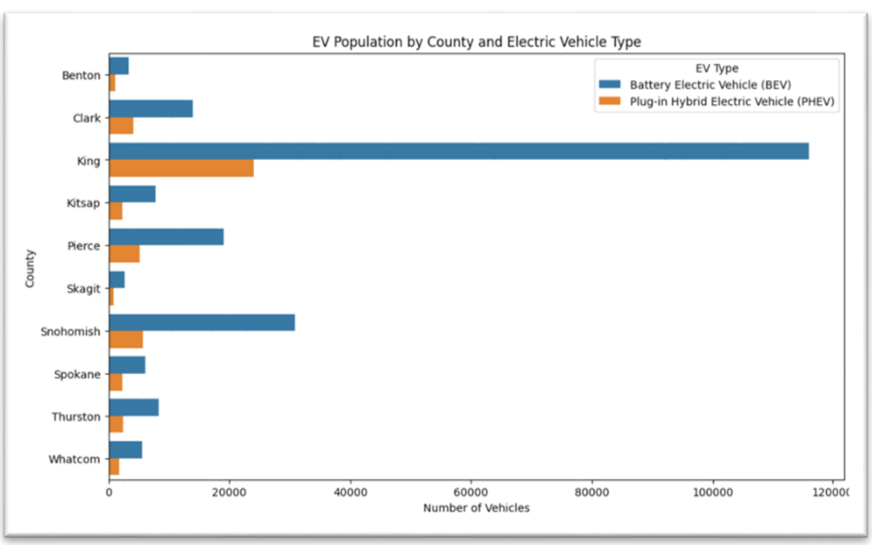
The visualization shows that Electric Range varies across model years and EV types. BEVs generally achieve higher electric ranges than PHEVs, particularly in newer model years, indicating an improvement in electric vehicle range over time.

2. Electric Range by Top EV Manufacturers and Vehicle Type



Electric Range varies considerably across manufacturers and vehicle types. BEVs generally show higher ranges, while PHEVs are concentrated at lower electric-range values. Tesla and Chevrolet show notable variation in BEV ranges.

3. EV Population by County and Electric Vehicle Type



King County has the highest EV population among the counties shown, with BEVs significantly outnumbering PHEVs. Other counties such as Snohomish and Pierce also show a strong presence of BEVs.

6. Key Insights & Findings

Summary of Findings

This project analyzed the **Electric Vehicle Population Data** using Python to understand the distribution, characteristics, and geographical patterns of electric vehicles. The analysis shows that **Battery Electric Vehicles (BEVs) have a much higher representation than Plug-in Hybrid Electric Vehicles (PHEVs)**. The dataset is mainly concentrated around recent model years, with **2023 being the most frequently occurring model year**. Electric Range shows considerable variation, with a mean of **37.40 miles** and a median of **0.00 miles**. The analysis also shows that EV population is geographically concentrated, with **King County having the highest overall vehicle count**, followed by Snohomish and Pierce counties.

Five Key Insights

• **BEVs dominate the dataset** — Battery Electric Vehicles have a much higher representation than Plug-in Hybrid Electric Vehicles.

• **Recent model years are strongly represented** — The mean model year is **2022.26**, while **2023** is the most frequently occurring model year.

• **Electric Range has high variability** — The mean Electric Range is **37.40 miles**, while the standard deviation is **76.74 miles**, showing substantial variation among vehicles.

• **EV adoption varies geographically** — King County has the highest EV population, followed by Snohomish and Pierce counties.

• **Electric vehicle characteristics vary across manufacturers** — The multivariate analysis shows differences in Electric Range across manufacturers and between BEVs and PHEVs.

Types of Analysis Applied

Type	Question Answered	Finding
Descriptive	What does the EV dataset look like?	BEVs dominate the dataset, recent model years are more common, and Electric Range varies considerably.
Statistical	How are the numerical variables distributed?	Electric Range shows the greatest variability, while Model Year is more concentrated around recent years.
Univariate	What is the distribution of individual EV characteristics?	EV types, manufacturers, model years, Electric Range, and CAFV eligibility show different distributions.
Bivariate	How are two EV characteristics related?	EV characteristics vary across counties, EV types, and other selected variables.
Multivariate	How do multiple EV characteristics interact?	Model Year, Electric Range, EV Type, Manufacturer, and County together reveal clearer patterns in EV adoption and vehicle characteristics.

7. Recommendations / Decision Support

• **Focus EV infrastructure planning on high-adoption counties.** Since EV population is concentrated in counties such as King, Snohomish, and Pierce, charging infrastructure can be prioritized in these areas.

• **Consider BEV dominance when planning infrastructure.** Since BEVs form the majority of the dataset, charging facilities should account for the higher presence of fully battery-electric vehicles.

• **Use Electric Range information for charging planning.** The large variation in Electric Range indicates that different EV users may have different charging requirements.

• **Consider manufacturer and EV type differences.** Variations in Electric Range across manufacturers and between BEVs and PHEVs can support better market and infrastructure planning.

• **Use recent model-year trends for future planning.** The strong representation of newer vehicles indicates the importance of considering continued EV adoption when planning future infrastructure.

8. Future Enhancements

• Use a larger and more regularly updated EV dataset to analyze current and future EV adoption trends.

• Develop an **interactive dashboard** using tools such as Power BI or Streamlit to allow users to explore EV population, manufacturers, counties, model years, and Electric Range.

• Extend the analysis with additional statistical tests to identify stronger relationships between EV characteristics.

• Develop a **predictive model** in future work to estimate EV adoption or Electric Range using relevant vehicle and geographical features.

• Perform more detailed geographical analysis using mapping and visualization techniques to identify EV adoption hotspots and support charging-infrastructure planning.

9. Conclusion

The **Electric Vehicle Population Data Analysis Using Python** project provides meaningful insights into the distribution and characteristics of electric vehicles. The analysis shows that **BEVs dominate the dataset**, recent model years have stronger representation, and Electric Range has substantial variation across vehicles. Geographical analysis indicates that EV adoption is concentrated in specific counties, particularly **King County, Snohomish, and Pierce**. Statistical analysis, visualization, and multivariate analysis together provide a clearer understanding of EV population patterns and can support **EV infrastructure planning, market analysis, and data-driven decision-making**.

10. Tools and Technologies Used

Programming Language: Python

Environment: Google Colab

Libraries Used:

- Pandas
- NumPy
- Matplotlib
- Seaborn

Data Source: Data.gov

Visualization: Matplotlib and Seaborn

Analysis Techniques: Data Cleaning, Data Transformation, Feature Engineering, Statistical Analysis, Univariate Analysis, Bivariate Analysis, and Multivariate Analysis.

11. Project Workflow

Data Collection → Data Cleaning → Data Transformation → Feature Engineering → Statistical Analysis → Data Visualization → Univariate Analysis → Bivariate Analysis → Multivariate Analysis → Insights & Findings → Recommendations → Future Enhancements

End of Documentation — Electric Vehicle Population Data Analysis Using Python Project